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--Manuscript Draft--

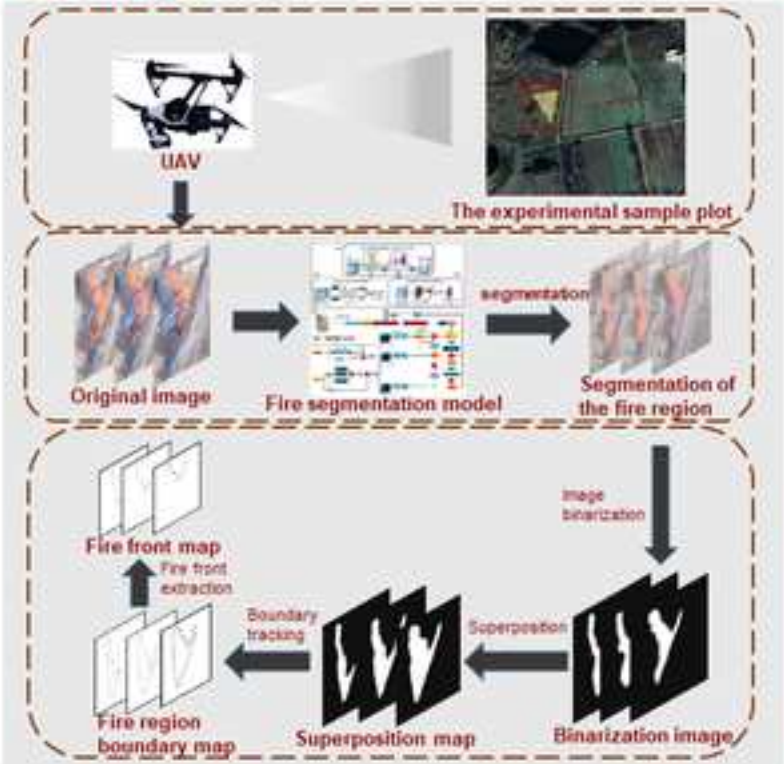
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Abstract:	The acceleration of global warming and intensifying global climate anomalies have led to a rise in the frequency of wildfires. However, most existing research on wildfire fields focuses primarily on wildfire identification and prediction, with limited attention given to the intelligent interpretation of detailed information, such as fire front within fire region. To address this gap, advance the analysis of fire front in UAV-captured visible images, and facilitate future calculations of fire behavior parameters, a new method is proposed for the intelligent segmentation and fire front interpretation of wildfire regions. This proposed method comprises three key steps: deep learning-based fire segmentation, boundary tracking of wildfire regions, and fire front interpretation. Specifically, the YOLOv7-tiny model is enhanced with a Convolutional Block Attention Module (CBAM), which integrates channel and spatial attention mechanisms to improve the model's focus on wildfire regions and boost the segmentation precision. Experimental results show that the proposed method improved detection and segmentation precision by 3.8% and 3.6%, respectively, compared to existing approaches, and achieved an average segmentation frame rate of 64.72 Hz, which is well above the 30 Hz threshold required for real-time fire segmentation. Furthermore, the method’s effectiveness in boundary tracking and fire front interpreting was validated using an outdoor grassland fire fusion experiment’s real fire image data. Additional tests were conducted in southern New South Wales, Australia, using data that confirmed the robustness of the method in accurately interpreting the fire front. The findings of this research have potential applications in dynamic data-driven forest fire spread modeling and fire digital twinning areas. The code and dataset are publicly available at https://github.com/makemoneyokk/fire-segmentation-interpretation.git.

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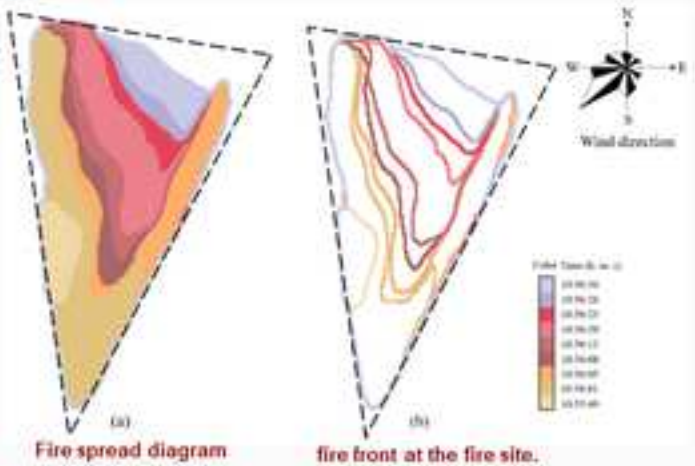
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METHODS



OUTCOME



CONCLUSION

Experimental results demonstrate that the proposed method improves detection and segmentation precision by 3.8% and 3.6%, respectively, compared to existing approaches, while achieving an average segmentation frame rate of 64.72 Hz, well above the 30 Hz threshold required for real-time fire segmentation. the method's effectiveness in boundary tracking and fire front interpreting was validated through an outdoor grassland fire fusion experiment's data.

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Intelligent Segmentation of Wildfire Region and Interpretation of Fire Front in Visible Light Images from the Viewpoint of an Unmanned Aerial Vehicle (UAV)

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ABSTRACT

The acceleration of global warming and intensifying global climate anomalies have led to a rise in the frequency of wildfires. However, most existing research on wildfire fields focuses primarily on wildfire identification and prediction, with limited attention given to the intelligent interpretation of detailed information, such as fire front within fire region. To address this gap, advance the analysis of fire front in UAV-captured visible images, and facilitate future calculations of fire behavior parameters, a new method is proposed for the intelligent segmentation and fire front interpretation of wildfire regions. This proposed method comprises three key steps: deep learning-based fire segmentation, boundary tracking of wildfire regions, and fire front interpretation. Specifically, the YOLOv7-tiny model is enhanced with a Convolutional Block Attention Module (CBAM), which integrates channel and spatial attention mechanisms to improve the model's focus on wildfire regions and boost the segmentation precision. Experimental results show that the proposed method improved detection and segmentation precision by 3.8% and 3.6%, respectively, compared to existing approaches, and achieved an average segmentation frame rate of 64.72 Hz, which is well above the 30 Hz threshold required for real-time fire segmentation. Furthermore, the method's effectiveness in boundary tracking and fire front interpreting was validated using an outdoor grassland fire fusion experiment's real fire image data. Additional tests were conducted in southern New South Wales, Australia, using data that confirmed the robustness of the method in accurately interpreting the fire front. The findings of this research have potential applications in dynamic data-driven forest fire spread modeling and fire digital twinning areas. The code and dataset are publicly available at https://github.com/makemoneyokk/fire-segmentation-interpretation.git.

1. Introduction

Global warming has led to the frequent occurrence of extreme weather events (Ning et al., 2024) such as heat waves, heavy rainfall, droughts, and localized climate drying (Chang et al., 2024). This has increased the flammability of vegetation (Park et al., 2022) and contributed to a rise in the frequency of wildfires (Kang et al., 2023; Wang and Zhang, 2020; Zhao et al., 2024). Due to the inherent unpredictability and instability of wildfires (Sousa et al., 2020), they are influenced by various factors, including climate, wind direction, and terrain slope (Zhang et al., 2023; Ribeiro et al., 2023). Once a wildfire occurs, it becomes extremely difficult to extinguish. As wildfires reach a certain scale, their destructive power, rate of spread, and suppression costs increase exponentially (Li et al., 2020; Lin et al., 2024), making fire control increasingly challenging. Therefore, timely intervention to slow the fire's rate of spread is crucial (Yar et al., 2023). During suppression efforts, firefighters typically assess the situation and control the fire based on the fire front. Thus, intelligent interpretation of the wildfire front's progression is essential for assisting firefighters in understanding the fire scene, making rapid decisions (De Almeida Pereira et al., 2021), and conducting suppression work efficiently.

The wildfire front marks the boundary between the fire region and unburned regions, representing the leading edge of wildfire spread. The interpretation of wildfire front data using computer vision and artificial intelligence algorithms is crucial in dynamic data-driven forest fire spread models and fire digital twins. However, it is challenging due to the hazardous nature of wildfires, the complexity of their physical and chemical mechanisms, and interference from background elements resembling fire (J. Li et al., 2023).

After the occurrence of forest fires, three primary methods of fire observation are available: satellites equipped with cameras, unmanned aerial vehicles (UAVs), and ground-based watchtowers (Allison et al., 2016) Satellite observations occur at altitudes of 30-100 kilometers, covering areas between 10-1000 square kilometers. While satellites offer extensive coverage due to their high vantage point, they suffer from significant drawbacks, including image update intervals of 1-2 days (Thangavel et al., 2023), high operational costs, and susceptibility to cloud interference in the data (Park et al., 2022). In contrast, UAVs and fixed cameras provide closer, continuous monitoring of wildfires, with observation distances of less than 4 kilometers and coverage areas of less than 50 square kilometers. These methods also yield higher-resolution images. However, fixed cameras are limited by their static positioning and fixed angles, making comprehensive observation of small-scale fires difficult. UAVs, on the other hand, can adjust flight paths, altitudes, and orientations according to real-time application scenarios (Boroujeni and Razi, 2024). Their flexible positioning and adaptable angles make UAVs more suitable for real-time wildfire monitoring, particularly in scenarios requiring high timeliness.

Wildfire research images captured by UAVs can be categorized into visible light (RGB) and thermal infrared (TIR) images (Li et al., 2023). TIR images, obtained through specialized sensors (Meng et al., 2022), provide information about the heat flux of objects (Oliver et al., 2022), but they suffer from low resolution (Boroujeni and Razi, 2024). Commonly used TIR image resolutions in recent years include 640x512, and the highest resolution of 1280x1024, with high acquisition costs (Almeida et al., 2023). For large forest fire areas covering tens of hectares, the precision of these images may not meet the needs of real practical applications. In contrast, RGB images can be captured at much higher resolutions, such as

4K or 8K, at a lower cost. While there are methods that fuse TIR and RGB images in wildfire research (Boroujeni and Razi, 2024), this fusion increases the complexity of system integration and the computational load of algorithm processing. Thus, UAV-based wildfire observation and fire information extraction using RGB images are more suitable for various application scenarios due to the lower operational costs and better quality of the captured image data (Cheng et al., 2024).

Rapidly interpreting detailed fire field information, such as fire spread rate, spread direction, and intensity from visible light images can enhance the efficiency of fire suppression efforts and deepen the understanding of wildfire spread mechanisms. The prerequisite for acquiring such information is achieving high-precision fire front interpretation. Currently, research on the wildfire front primarily relies on thermal sensors and spread models. Methods relying on thermal sensors ; Pastor et al., 2006) involve high image acquisition costs and fail to meet the requirements of practical applications. On the other hand, fire front extracted using spread models (Hernández Encinas et al., 2007; Zhai et al., 2020) do not represent actual fire data and therefore cannot accurately reflect real fire conditions. To address these limitations, visible light image-based methods, integrating deep learning with traditional mathematical algorithms, offer a promising solution for fire front interpretation. This approach involves applying wildfire segmentation algorithms and fire front interpretation methods to extract fire regions from images, followed by interpreting the wildfire front. However, limited research has been conducted on this approach.

To improve wildfire front interpretation and address gaps in intelligent wildfire front analysis using UAV-acquired RGB images, we propose a new method for the intelligent segmentation of wildfire regions and the interpretation of fire field fronts from these images. This method enables the interpretation of prairie fire front and consists of two main components: fire region segmentation and fire field front interpretation. The process begins with preprocessing the image data from outdoor prairie fire fusion experiments, which includes image registration and enhancement. Next, a fire segmentation model based on deep learning is trained and optimized. The optimized model is then used to segment the preprocessed images, binarizing the identified fire regions. Subsequently, a region-locking-based wildfire front interpretation algorithm is applied to analyze the segmented fire regions, focusing on the region of interest within the image. In this process, the hybrid intelligent interpretation of the wildfire front is expected to be achieved through an improved YOLOv7-tiny-based fire segmentation model and a region-locking wildfire front interpretation algorithm. The key contributions of this research are as follows:

1. We introduce the Convolutional Block Attention Module (CBAM) and propose a fire segmentation model based on the YOLOv7 lightweight network (C.-Y. Wang et al., 2023) by utilizing the CBAM module, which combines the channel and spatial attention mechanisms, in place of the traditional convolutional operation after the feature splicing layer. The performance of the proposed model is evaluated using metrics such as precision, F1-score, Average Precision (AP), Parameters, Floating Point Operations Per Second (FLOPs), and Frames Per Second (FPS).

2. An intelligent wildfire front interpretation algorithm based on region-locking has been proposed to achieve an accurate interpretation of the wildfire front. This algorithm applies boundary tracking and region-locking techniques to the region of interest in fire images, facilitating the subsequent extraction of wildfire behavior.

3. To the best of our knowledge, this is the first study to utilize UAV-acquired RGB images for intelligent wildfire front interpretation by combining a deep learning-based fire segmentation model with a region-locking-based fire front extraction algorithm. This hybrid approach integrates deep learning methods with traditional mathematical algorithms to achieve high-precision wildfire front interpretation.

The remainder of this paper is organized as follows: Section II provides an overview of the related work. Section III presents the proposed novel wildfire segmentation model based on an enhanced YOLOv7-tiny network and a region-locking-based wildfire front interpretation algorithm. Section IV evaluates the performance of the proposed wildfire segmentation model and validates the proposed front interpretation algorithm using real outdoor prairie fire fusion experimental image data. Finally, Section V concludes the study and offers recommendations for future research.

2. Related work

Due to the hazardous nature of wildfires, the complexity of their physical and chemical mechanisms, and interference from background elements resembling fire, the research on the intelligent interpretation of wildfire front has been facing certain challenges. To realize the intelligent interpretation of wildfire regions in visible images under UAV view involves: an efficient real-time fire region segmentation model and wildfire front intelligent interpretation technology. In this section, we review the related work of the above two technologies and analyze them accordingly.

2.1. Efficient real-time fire region segmentation model

Traditional fire segmentation algorithms primarily rely on the visual features of fire, such as color (Chen et al., 2024), texture, shape (P. Wang et al., 2023), fire flickering variations (Ko et al., 2010), and spatial color variations (Cai et al., 2020), to extract fire regions in complex backgrounds (Chen et al., 2021). However, these traditional algorithms are prone to false detection when encountering objects with similar color, texture, shape, and motion patterns to fire (Yuchen et al., 2023). Additionally, their image processing speed is slow. Thus, it is crucial to explore efficient methods for fire segmentation.

With the advent of machine learning, clustering algorithms have been applied to fire segmentation (Shama et al., 2023), leading to improvements in segmentation efficiency. These methods classify pixels by finding a k-cluster division scheme through iterative processes. For instance, researchers (Shama et al., 2023) have utilized multi-scale segmentation algorithms to generate targets based on datasets with polarization and texture features. Subsequently, the k-means clustering algorithm was applied to classify image pixels, yielding promising segmentation results. However, the performance of such methods heavily depends on the selection of the k-value, which affects the segmentation results. Furthermore, these methods exhibit limited robustness against interference from fire-like objects.

Both traditional fire segmentation algorithms and clustering algorithms perform well in specific scenarios. However, their false detection rates increase significantly when dealing with objects that share similar visual features to fire (Chen et al., 2024). Additionally, these methods often involve complex mathematical computations, resulting in low computational efficiency. These limitations have driven the development of deep learning-based fire segmentation methods.

Deep learning methods have made breakthroughs in the field of image segmentation, especially Convolutional Neural Networks (CNNs) and their variants are widely used in fire segmentation tasks. These methods automatically extract multi-scale fire features through end-to-end training, and the anti-interference ability for fire-like objects has been improved, while the efficiency and accuracy of wildfire segmentation have also been improved. Researchers have implemented various optimizations to deep learning frameworks. For instance, lightweight models like MobileNetV3 have been incorporated into the DeepLabV3+ network to reduce computational complexity and enhance real-time performance (Li et al., 2022). Improvements to the U-Net model, such as integrating a transformer to maintain accuracy while reducing parameters, have also been explored (Shahid et al., 2023). Other approaches include combining EfficientNet-B5 and DenseNet-201 with visual transformers (TransUNet and TransFire) and the EfficientSeg model to enhance wildfire detection and precisely delineate wildfire regions (Ghali et al., 2022). Dual fusion attention residual feature attention (DARA) has been proposed within the DeepLabV3+ framework to enhance fire feature extraction and achieve precise identification (Liu et al., 2023). The CBAM attention mechanism is introduced based on deeplabV3+ network architecture to fully extract

the image features to suppress the background interference and so on (Chen et al., 2023). However, balancing segmentation accuracy and real-time performance is most effectively addressed by the YOLO (You Only Look Once) series of networks (Wang et al., 2022). YOLO (Redmon et al., 2016) integrates target detection and segmentation tasks, demonstrating significant potential in real-time segmentation scenarios due to its fast inference capabilities.

YOLO is a single-stage target detection network, with its core concept being the transformation of the target detection task into a regression problem to enable end-to-end prediction with high real-time performance. From YOLOv1 (Hao et al., 2024) to YOLOv8 (L. Li et al., 2023), the network structure has been continuously optimized, and the performance has been significantly improved. The YOLOv1-v3 networks improve the performance of small target recognition through multi-scale detection. Among them, YOLOv3 (L. Zhao et al., 2022) has the ability of multi-scale detection. Zhao et al. (2022) replaced Darknet53, the feature extraction network in YOLOv3, with EfficientNet to enhance the model's ability to detect small targets. While this approach significantly improved small-target detection, the overall detection accuracy remains suboptimal. YOLOv4 (S. Zhao et al., 2022) introduced CSPNet and the Mish activation function within the Darknet framework, achieving higher detection accuracy and speed. Zhao et al. (2022) improved YOLOv4 by replacing the Path Aggregation Network (PANet) with a simplified weighted bidirectional feature pyramid network (Bi-FPN) as the feature fusion network. By incorporating dynamic fire characteristics, this method accounted for the multidimensional nature of fire and further enhanced detection accuracy. However, YOLOv4 requires specific dependency environments and is complex to deploy. YOLOv5 (He et al., 2024) shifted to the PyTorch framework, using a lightweight backbone network and simple training strategies to further improve model flexibility and inference speed. YOLOv5 became an important foundation for real-time segmentation due to its lightweight design and efficiency. Subsequently, YOLOv7 (Wang et al., 2022) further optimizes the network structure to meet the demand of high real-time performance, the complexity of the network structure is increased but the detection ability and detection speed are also further improved, and the model achieves a good balance between detection speed and accuracy.YOLOv8 (L. Li et al., 2023) supports the design of the anchorless frame, which improves the detection ability of targets of different sizes, shapes and poses, and strengthens the generalization ability of the model, but the real-time performance is a bit lower than that of YOLOv7.

Experiments on visible and thermal image datasets demonstrate that the YOLO-based segmentation model excels in real-time performance and accuracy, meeting real-time application requirements (Yandouzi et al., 2024). With the introduction of the attention mechanism and multi-scale feature fusion, the segmentation accuracy values outperform deep learning methods such as H-Net and U-Net (Cao et al., 2023). These advantages have motivated the application of the YOLO segmentation model to the segmentation task of wildfire series image data, for example, researchers (Niu et al., 2023) proposed a forest fire segmentation model (FFDSM) based on infrared images from UAVs, which efficiently recognizes the fire region obscured by vegetation by integrating YOLOv5s-seg, ECA module, and SPPFCSPC module to achieve a high segmentation accuracy. However, the dependence of infrared images on expensive sensor devices limits the popularization of its real-world applications. To address this problem, another study (Cao et al., 2023) proposed the segmentation detection algorithm YOLO-SF based on the YOLOv7-tiny network, while creating a fire segmentation dataset (FSD). The algorithm introduces the MobileViTv2 module and CBAM module, which improves the localization ability and segmentation accuracy through the varifocal Loss function. Although the method achieves high wildfire segmentation performance while keeping the network lightweight, there is still room for further improvement in segmentation accuracy.

In order to meet the practical demands for real-time, high segmentation accuracy and low model complexity in firefighting, this paper proposes an improved fire segmentation model based on the YOLOv7-tiny network. The model achieves accurate segmentation of wildfires in a resource-constrained environment by optimizing the network architecture. On the basis of wildfire segmentation, another important task to realize intelligent interpretation is the wildfire front interpretation technology.

2.2. Wildfire front intelligent interpretation technology

With the gradual application of computer vision technology to fire observation, people are no longer satisfied with the automatic detection of wildfires but want to obtain more detailed information about the fire, such as the fire spread rate, spread direction, intensity, etc. The prerequisite for the acquisition of this information is to realize the extraction of high-precision fire front. However, there are fewer research works on wildfire front interpretation, and most of them are manual visual interpretation or based on thermal sensors ; Pastor et al., 2006)and spread models ; Zhai et al., 2020).

Currently, the semi-automated interpretation of wildfire fronts can be categorized into three main categories: methods based on thermal sensor heat flow inversion, methods utilizing wildfire spread models to predict the fire front, and methods employing thermal image threshold-searching criteria to identify the fire front.

The methods based on thermal sensor heat flow inversion are to calculate the heat flux from the fire heat flow obtained from heat sensing measurements, and then to obtain the wildfire front location by the inverse method. For example, Chetehouna et al (2008) utilized specific thermal sensors to measure the heat flow from the fire in four horizontal directions, then obtained the heat flux through the approximate resolution of the radiative transfer equation, and finally the inverse method determined the fire front location. Although this type of method can provide real data on the fire front, it needs to deal with sensor data noise interference before finding the wildfire front with corresponding errors and high cost.

The wildfire spread model-based methods predict the fire front by simulating fire spread can be categorized into two types: methods based on cellular automata for simulating fire front spread and learning-based methods for wildfire spread prediction. Cellular automata-based methods, such as the model proposed by Hernández et al. (2007), simulate fire front dynamics by improving the Karafyllidis and Thanailakis models to represent circular fire spread. Learning-based models, like the one developed by Zhai et al. (2020) using a narrow-band level-set method, predict short-term wildfire spread. This type of spread model can generate wildfire front but is based on statistical and empirical rules, making its accuracy dependent on the quality of the rule design. Consequently, the predicted fire front cannot be considered true fire front data.

Thermal image thresholding methods can be further divided into channel threshold search and temperature matrix threshold search approaches. Channel threshold search methods involve processing thermal images to identify optimal channels, assigning thresholds for each channel, and locating the fire front by identifying the last white pixel in each column. For example, Chetehouna et al. (2008) used a channel matrix computed via a direct linear transformation (DLT) algorithm to determine the fire front's position within the propagation zone by identifying the optimal threshold. Similarly, threshold-based methods applied to thermal image temperature matrices use specific criteria to locate fire front. For instance, Pastor et al. (2006) employed the IR camera to generate a temperature distribution for each pixel and established a criterion to identify the fire front. While these methods provide accurate fire front data, they require high-quality sensors and need to be continually tested to obtain optimal thresholds and have poor generalization capabilities.

With the continuous improvement of fire segmentation technology in terms of accuracy and efficiency, the wildfire segmentation algorithm based on the YOLO network has been able to realize real-time processing. In this paper, a new fire front interpreting method is proposed based on

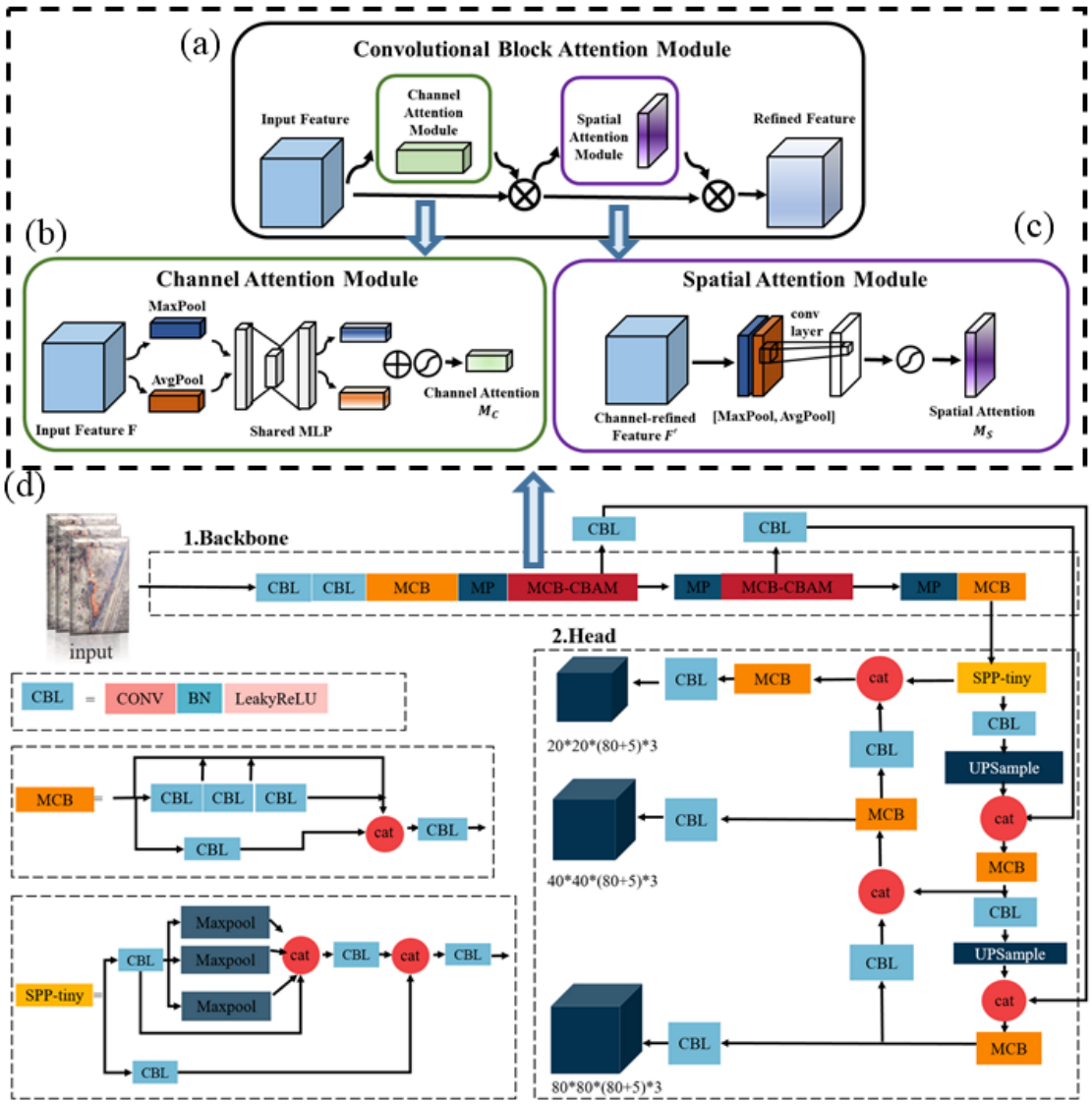


Fig. 1. Structure of our model. (a) Structure of CBAM Module. (b) Channel attention module. (c) Spatial attention module. (d) Improved Lightweight YOLOv7 Model.

YOLOv7-tiny wildfire segmentation algorithm, which extracts the fire front from a series of frames of images or videos through the computation process of fire region segmentation, edge detection analysis (Ma, 2024; Ma et al., 2024), and geometric topology. Compared to methods based on thermal sensors and spread models, this approach offers lower costs, higher computational efficiency, and better suitability for time-critical on-site fire front extraction tasks.

3. Method

3.1. Deep learning-based fire segmentation algorithm

3.1.1. *Fire segmentation model based on improved YOLOv7-tiny network*

YOLOv7 (C.-Y. Wang et al., 2023) is a one-stage algorithm known for its flexibility and versatility in practical applications. The YOLOv7-tiny network is a lightweight version of the original YOLOv7 architecture, optimized for deployment on embedded devices. While it is more compact than the full YOLOv7 network, the difference in segmentation precision between the two is minimal. To further enhance the segmentation accuracy of the YOLOv7-tiny model for fire detection, the CBAM (Woo et al., 2018) was integrated into the original framework. This module replaces the traditional convolutional operation after the feature splicing layer, constructing a more effective backbone network and improving the model's segmentation precision.

CBAM (Woo et al., 2018) is a convolutional attention module consisting of a Channel Attention Module and a Spatial Attention Module. The Channel Attention Module extracts two spatial feature descriptions by applying global average pooling and global maximum pooling to the input feature maps. These feature descriptions are then passed through a shared multilayer perceptron (MLP), where they are summed to generate the channel attention map. Concurrently, the Spatial Attention Module performs maximum pooling and average pooling across the channel dimension of the input feature maps, producing two single-channel feature maps that are concatenated and processed through a convolutional layer to generate the spatial attention map. The spatial attention map is then multiplied by the channel attention map to produce the final output feature map. This process adaptively assigns weights to each channel and spatial location, enhancing the model's segmentation precision. The structure of CBAM is illustrated in Fig. 1(a), with the Channel Attention and Spatial Attention sub-modules shown in Fig. 1(b) and Fig. 1(c), respectively.

In CBAM, input data is first processed into a feature map through a convolutional operation. During the channel attention phase, global maximum pooling and global average pooling are applied to the input feature maps, producing two vectors corresponding to the number of channels. These vectors are passed through a shared fully connected layer, where attention weights are adaptively assigned to each channel. The global maximum and average feature vectors are then combined to generate the final attention weight vector, which is multiplied with the original feature map to create a weighted channel feature map, emphasizing relevant channels while suppressing irrelevant ones. In the spatial attention phase, maximum pooling and average pooling are performed along the

spatial dimension to capture features at different context scales, which are then concatenated. The concatenated feature maps are processed by a convolutional layer to generate spatial attention weights, which are constrained between 0 and 1 using a Sigmoid activation function. These spatial attention weights are applied to the original feature map to emphasize important regions while diminishing less relevant areas. The output of the CBAM module can be seamlessly integrated into subsequent layers. Its lightweight design allows for a significant improvement in model performance without substantially increasing model complexity or computational cost.

The backbone of the original YOLOv7 consists of three primary modules: the convolutional layer (CBL: Conv2D + Batch Normalization + LeakyReLU activation), the MCB module, and the MaxPooling module. While the CBAM enhances performance when inserted at various points, applying it to high-resolution feature maps significantly increases the number of model parameters. In this research, the CBAM module replaces one of the traditional convolutional layers following the connection layer of multiple MCB modules to form the backbone network (as shown in Fig. 1(d)).

3.1.2. Evaluation metrics for fire segmentation

As the YOLOv7-tiny network is utilized to further segment objects based on the initial detection frame, both the Box and Mask outputs were evaluated. The evaluation metrics employed in this research include Precision, Recall, F1-score, and AP, which are widely used for performance assessment.

$$Precision = \frac{TP}{TP+FP}, \tag{1}$$

$$Recall = \frac{TP}{(TP+FN)}, \tag{2}$$

$$F1-score = \frac{2\times Precision\times Recall}{Recall+Precision}, \tag{3}$$

$$AP = \sum_n (R_n - R_{n-1}) P_n. \tag{4}$$

The TP, FP, and FN refer to true positives, false positives, and false negatives, respectively, for the target object (fire) and background in the detection and segmentation tasks. Specifically, TP represents the number of correctly detected/segmented wildfires, FP indicates the number of incorrectly detected/segmented wildfires, and FN reflects the number of undetected/segmented wildfires. These metrics are differentiated by detection and segmentation, often abbreviated as P(B), P(M), R(B), R(M), AP(B), AP(M), F1(B), and F1(M) for precision, recall, AP and F1-score of both the box and mask outputs. Additionally, R_n denotes the recall at the nth point, and P_n denotes the precision at the nth point. The corresponding mathematical expressions are presented in the above equations.

Precision refers to the proportion of predicted positive samples (fire) that are actually positive, representing the percentage of correctly identified fire regions among all regions predicted as fire. Recall, on the other hand, measures the proportion of actual positive samples that were correctly predicted, indicating the percentage of correctly identified fire regions among all true fire regions. The F1-score combines Precision and Recall to provide an overall assessment of the model's performance. AP is the average of precision values computed at varying recall levels, offering a comprehensive evaluation of the model's performance across different thresholds.

3.2. Fire front extraction algorithm based on region locking

(1) Boundary tracking

Boundary tracking (Suzuki and KeiichiA be, 1985) involves traversing image pixels and utilizing the hierarchical relationships between boundary pixels to delineate graphic boundaries. Suzuki introduced the concept of topological relationships among boundary pixels, employing containment relationships to trace the outline of shapes by labeling the pixels. Here it can be replaced with other edge detection algorithms, such as those using Canny, Sobel operator, Laplacian operator, Prewitt operator and Roberts operator.

First, the top row, bottom row, leftmost column, and rightmost column of an image form its frame. We denote the pixels located in row i and column j by row number and column number (i,j). The following coordinate system is used here: row number i is incremented from top to bottom; column number j is incremented from left to right. We define a three-channel image I, consisting of a red channel $R(i,j)$, a green channel $G(i,j)$, and a blue channel $B(i,j)$, which can be represented as:

$$\mathrm{I}(i,j) = \begin{bmatrix} R(i,j) \\ G(i,j) \\ B(i,j) \end{bmatrix}. \tag{5}$$

Here (i,j) denotes the pixel coordinates of the image. Graying of this three-channel image I yields the image L.

$$\mathrm{L}(i,j) = \frac{R(i,j)+G(i,j)+B(i,j)}{3}. \tag{6}$$

By binarizing the grayscale image L, the image is transformed according to the threshold value to get the binarized image A:

$$\mathrm{A}(i,j) = \begin{cases} 1 & if\ \ L(i,j) \geq T \\ 0 & if\ \ L(i,j) < T \end{cases}. \tag{7}$$

Where 1 denotes white as foreground pixels, 0 denotes black as background pixels, and T is a threshold value, usually set to 127. Suppose the resulting binarized image consists of a matrix of M×N pixels, where M and N are the height and width of the image, respectively. The boundary can be represented by an ordered set of points, $\mathrm{B} = \{(a_k, b_k) | k = 1,2,\dots,n\}$, where n is the number of boundary pixels and (a_k, b_k) denotes the kth pixel point on the boundary.

We first define two numbers, initial boundary number (NBD) and column number ($LNBD$). The NBD is 1, denoted as the first boundary. Since the object may have holes in the middle, we divide the object boundaries into outer and hole boundaries. Boundary number NBD is used to identify each newly detected boundary pixel, either an outer boundary or a hole boundary, and $-NBD$ denotes boundary pixels or hole boundary pixels that have already been processed. Meanwhile, the column number $LNBD$ is initialized to 1, which is used to determine the parent boundary of the boundary point, and is reset to 1 at the beginning of each new row. Here we determine the pixel according to the value of the pixel, the operation is carried out in three cases:

1. if the current pixel $\mathrm{A}(i,j) = 1$ and the left neighbor pixel $\mathrm{A}(i,j-1) = 0$, then the current pixel is the starting point of the outer boundary, update the value of NBD, $NBD{+}{=}\,1$, and make $(i_2,j_2) = (i,j-1)$, where $(i,j-1)$ is the position of the left pixel of the current pixel.

2. If the current pixel $\mathrm{A}(i,j) \geq 1$ and the left neighbor pixel $\mathrm{A}(i,j+1) = 0$, then the current pixel is the starting point of the hole boundary, update the values of NBD, $NBD{+}{=}\,1$, and make $(i_2,j_2) = (i,j+1)$, where $(i,j+1)$ is the position of the right pixel of the current pixel.

3. If none of the above conditions are met, determine whether the current pixel value $\mathrm{A}(i,j)$ is not equal to 1. If this condition is true, set $LNBD = |\mathrm{A}(i,j)|$ and continue the raster scan until reaching the bottom-right corner of the image.

Determine the parent boundary according to the type of the previous boundary $\mathrm{B}'(a_{k-1}, b_{k-1})$ and the current newly encountered boundary $\mathrm{B}(a_k, b_k)$. If both the current boundary $\mathrm{B}(a_k, b_k)$ and the previous boundary $\mathrm{B}'(a_{k-1}, b_{k-1})$ are outer boundaries or both the current boundary B and the previous boundary $\mathrm{B}'(a_{k-1}, b_{k-1})$ are hole boundaries, the current boundary is the parent boundary of the previous boundary $\mathrm{B}'(a_{k-1}, b_{k-1})$. If the current boundary $\mathrm{B}(a_k, b_k)$ and the previous boundary $\mathrm{B}'(a_{k-1}, b_{k-1})$, one is a hole boundary and the other is an outer boundary, then the parent boundary is still the previous boundary $\mathrm{B}'(a_{k-1}, b_{k-1})$.

Starting from the boundary starting point (i_2,j_2), with (i,j) as the center, find the first non-zero pixel point (i_1,j_1) of 4 or 8 neighborhoods in a clockwise direction, and this pixel will then be used as the starting point for the subsequent search in the counterclockwise direction. If find

non-zero pixel (i_1, j_1), then make $(i_2, j_2) = (i_1, j_1)$, $(i_3, j_3) = (i, j)$. Otherwise, directly assign $-NBD$ to the current pixel point, i.e., make $\mathrm{A}(i, j) = -NBD$, determine whether the gray value $\mathrm{A}(i, j) \neq 1$ of the current pixel point is valid or not, and if it is valid, then make $LNBD = |\mathrm{A}(i, j)|$ and continue the raster scanning until the scanning is over when it reaches the vertex in the lower right corner of the image.

Next, with (i_3, j_3) as the center and (i_2, j_2) as the starting point, search counterclockwise within the 4- or 8-neighborhood for the first non-zero pixel (i_4, j_4) to expand the loop. If $\mathrm{A}(i_3, j_3 + 1) = 0$, set $\mathrm{A}(i_3, j_3) = -NBD$ and add (i_3, j_3) to the set of boundary points B, i.e., set $(a_k, b_k) = (i_3, j_3)$ and increment k. If $\mathrm{A}(i_3, j_3 + 1) \neq 0$, and $\mathrm{A}(i_3, j_3) = 1$, set $\mathrm{A}(i_3, j_3) = NBD$, and similarly, add$(i_3, j_3)$to the set of boundary points B, i.e., set $(a_k, b_k) = (i_3, j_3)$, and increment k by 1. In all other cases, leave $\mathrm{A}(i_3, j_3)$ unchanged.

Subsequently, determine whether the contour has returned to the origin. If $(i_4, j_4) = (i, j)$ and $(i_3, j_3) = (i_1, j_1)$, this indicates a return to the origin. At this point, check the validity of the current pixel value $\mathrm{A}(i, j) \neq 1$. If valid, update $LNBD = |\mathrm{A}(i, j)|$ and continue the raster scan until it reaches the bottom-right corner of the image. Otherwise, update $(i_3, j_3) = (i_4, j_4)$ and $(i_2, j_2) = (i_3, j_3)$, and continue searching in the counterclockwise direction for the next non-zero pixel, repeating this process until the current pixel is a checked point, i.e., $|\mathrm{A}(i_3, j_3)| = NBD$. At that point, exit this loop and re-evaluate which of the previously mentioned three cases the grayscale value of the current pixel belongs to.

Finally, determine whether $\mathrm{A}(i, j) \neq 0$, is valid, and if it is valid, then let $LNBD = |\mathrm{A}(i, j)|$, and continue raster scanning until the scanning ends when it reaches the lower right vertex of the image. At the end of scanning all the pixel points labeled as boundary pixels form the boundary pixel point set $\mathrm{B} = \{(a_k, b_k) | k = 1,2, \ldots, n\}$. Thus the boundary of the input image is obtained(Suzuki and KeiichiA be, 1985). For a detailed explanation, please refer to Algorithm 1.

(2) Region locking to extract the fire front

Based on the boundaries obtained from the above tracking, we use several pixel points to determine a polygonal region of interest, here three-pixel vertices of the region of interest P_1, P_2 *and* P_3 are utilized to obtain a triangle$\Delta P_1 P_2 P_3$ to construct the triangular region T:

$$\mathrm{T} = \{(x, y) \in R^2 | (x, y) \in \Delta P_1 P_2 P_3\}. \quad (8)$$

Create a blank mask map M with the same size as the boundary image, initially with background color C_{bg}:

$$C_{bg} = (0,0,0), \quad (9)$$

$$\mathrm{M}(i, j) = C_{bg}, \forall (i, j) \in [0, W) \times [0, H). \quad (10)$$

Here H is the height of the image and W is the width of the image. Iterate over each pixel (x, y) of the image I to determine whether it is within the triangle region T. The steps are as follows:

Know the order of the vertices of the triangle as (P_1, P_2, P_3), initialize the intersection counter for a given point (V_x, V_y), and set count=0. Draw a ray horizontally to the right from a given point (V_x, V_y). Traverse the three sides of the triangle: for each side (P_1, P_2), where $P_1 = (x_i, y_i)$and$P_2 = (x_{i+1}, y_{i+1})$, determine the following:

1. Determine whether the point is on a horizontal edge: if $V_y = y_i$and $x_i \leq V_x \leq x_{i+1}$ or $V_y = y_{i+1}$ and $x_i \leq V_x \leq x_{i+1}$, it means that the given point is exactly on the horizontal line of a certain edge and between the two endpoints of the edge, then the point is determined to be on a horizontal edge.

2. Determine whether an edge crosses a ray: if $(y_i > V_y) \neq (y_{i+1} > V_y)$, then it means that one of the vertices of an edge is on top of the ray and the other vertex is on the bottom of the ray, then determine that the edge crosses the horizontal ray, and compute the intersection of the edge and the ray, $x_{intersect}$:

$$x_{intersect} = x_i + \frac{(V_y - y_i) \cdot (x_{i+1} - x_i)}{y_{i+1} - y_i}. \quad (11)$$

3. Determines whether the point is inside or outside the triangle: According to the intersection counter count to determine the location of the point, if the final intersection counter count is an odd number, then the point is inside the triangle; if it is an even number, then the point is outside the triangle. Here the triangle is replaced by a polygon in the same way.

When the pixel point is inside the triangle or on the boundary, the original image pixel value is kept, otherwise, it is set to the background color, the mathematical expression is as follows:

$$\mathrm{M}(\mathrm{x}, \mathrm{y}) = \begin{cases} I(x, y), & if \ \text{count} \in \mathrm{O}, \mathrm{O} = \{\mathrm{x} | \mathrm{x} = 2\mathrm{k} + 1, \mathrm{k} \in \mathrm{Z}\} \\ C_{bg}, & if \ \text{count} \in \mathrm{E}, \mathrm{E} = \{\mathrm{x} | \mathrm{x} = 2\mathrm{k}, \mathrm{k} \in \mathrm{Z}\} \end{cases}. \quad (12)$$

This results in a mask map $\mathrm{M}(x, y)$, with $\mathrm{M}(x, y)$ being the mask image containing the wildfire front.

Algorithm 1. Boundary tracking

Input: image A=$\{\mathrm{A}(i, j)\}$, image A consists of a matrix of M*N pixels
Output: $\mathrm{B} = \{(a_k, b_k) | k = 1,2, \ldots, n\}$

$NBD = 1, LNBD = 1, k = 1$
for $i = 0 \rightarrow$ M **do**
  for $j = 0 \rightarrow$ N **do**
    if $\mathrm{A}(i, j) == 1$ and $\mathrm{A}(i, j-1) == 0$ **then**
      $NBD += 1, (i_2, j_2) \leftarrow (i, j-1)$
    elseif $\mathrm{A}(i, j) \geq 1$ and $\mathrm{A}(i, j+1) == 0$ **then**
      $NBD += 1, (i_2, j_2) \leftarrow (i, j+1)$
    else
      goto step 35
    end if
    if $\mathrm{B}(a_k, b_k)$ and $\mathrm{B}'(a_{k-1}, b_{k-1})$ are either the same hole boundary or outer boundary **then**
      the parent boundary of $\mathrm{B}(a_k, b_k)$ is the parent boundary of $\mathrm{B}'(a_{k-1}, b_{k-1})$
    else
      the parent boundary of $\mathrm{B}(a_k, b_k)$ is $\mathrm{B}'(a_{k-1}, b_{k-1})$
    end if
    find the 4(8) neighbors of (i, j) clockwise with (i, j) as the center and (i_2, j_2) as the starting point.
    if there are non-zero pixel points(i_1, j_1) **then**
      $(i_2, j_2) \leftarrow (i_1, j_1), (i_3, j_3) \leftarrow (i, j)$
    end if
    repeat
    with (i_3, j_3) as the center and (i_2, j_2) as the start point look counterclockwise for the first non−zero pixel point in the 4(8) neighborhood of (i_3, j_3) defined as (i_4, j_4)
      if $\mathrm{A}(i_3, j_3 + 1) == 0$ **then**
        $\mathrm{A}(i_3, j_3) \leftarrow -NBD, (a_k, b_k) = (i_3, j_3), k++$
      elseif $\mathrm{A}(i_3, j_3 + 1) \neq 0$ and $\mathrm{A}(i_3, j_3) == 1$ **then**
        $\mathrm{A}(i_3, j_3) \leftarrow NBD, (a_k, b_k) = (i_3, j_3), k++$
      else
        $\mathrm{A}(i_3, j_3) = \mathrm{A}(i_3, j_3)$
      end if
      if $(i_4, j_4) == (i, j)$ and $(i_3, j_3) == (i_1, j_1)$
        goto step 35
      else
        $(i_2, j_2) = (i_3, j_3)$ and $(i_3, j_3) = (i_4, j_4)$
      end if
    until $|\mathrm{A}(i_3, j_3)| == NBD$
    if $\mathrm{A}(i, j) \neq 1$ **then**
      $LNBD \leftarrow |\mathrm{A}(i, j)|$
    end if
    $j += 1$
  end for
  $LNBD = 1$
  $i += 1$
end for

4. Experiments and discussion

4.1. Fire segmentation experiment

4.1.1 . Dataset description

For deep learning-based network models, both the quality and quantity of image datasets, as well as the accuracy of labeled data, directly impact model performance. A significant limitation in fire segmentation research is the scarcity of labeled image data. To address this issue and provide an adequate number of fire images for the network model, this research utilized the publicly available Fire-Segmentation-Dataset (FSD) (Yixuan Su ,2023)as the data source for training the fire segmentation model.

The image dataset comprises a total of 5,000 images, including 3,203 fire images and 1,797 non-fire images. The dataset was labeled and split into training and test sets at a 9:1 ratio. Additionally, 10% of the training set was reserved for validation. Consequently, the proportions of the training, test, and validation sets are approximately 8:1:1, as detailed in Table 1.

Table 1

Introduction of the dataset.

Data partition	Wildfire	Non-wildfire	All
Train	2595	1455	4050
Test	326	174	500
Val	282	168	450
All	3203	1797	5000

The dataset includes a range of negative instance images such as sunsets, city lights, street lights, and headlights, with a sample of these negative instances depicted in Fig. 2(a). It encompasses a variety of scenes, including grasslands, buildings, kitchens, mountains, cars, and other settings. The dataset features high-speed photography from multiple perspectives, including flat, top, and bird's-eye views. Fig. 2(b) presents some sample images from the dataset along with their corresponding labels.

The fire scene image dataset includes images of various resolutions, such as 550 × 374, 440 × 587, and 385 × 385. These images are uniformly resampled to 640 × 640 for model input during processing. This resampling ensures consistent model design without impacting experimental results. After segmentation, images are restored to their original size. Fig. 3 illustrates the quantity, spatial distribution, and size of image data labels, showing that the dataset contains approximately 6,000 fire labels. The distribution of labeled data in the images exhibits both uniformity and diversity.

4.1.2 Implementation details

The model was implemented using PyTorch and trained on a system running Windows 10 with an NVIDIA GeForce RTX 3090 GPU featuring 24 GB of RAM. Details of the experimental environment are provided in Table 2. YOLOv7 was utilized as the backbone network, with an input image size of 640 × 640. The model was trained for 300 epochs with a batch size of 4. Stochastic Gradient Descent (SGD) was employed as the optimization solver, with a mosaic value of 1.0, momentum set to 0.937, and weight decay of 0.0005. The initial learning rate was 0.01, and an exponential decay method was applied to reduce the learning rate gradually.

Table 2

Experimental environment.

Schedule	Capacity
Parameters Configuration	Operating system Windows10
CPU	Intel(R) Core(TM) i7-8700K CPU@ 3.70GHZ
GPU	NVIDIA GeForce RTX 3090
RAM	24GB
Deployment Environment	Python3.8.19
Deep learning framework	PyTorch1.13.1
Accelerated Computing Architecture	CUDA11.7

4.1.3. Performance comparison

We analyzed the performance of our method in comparison with wildfire segmentation methods based on various YOLO series networks, including YOLOv5, YOLOv7, YOLOv8, and YOLOv9 with different network architectures. To make a fair comparison, we used the same hyperparameter settings in training, and the corresponding evaluation results are given for the above-mentioned network models, summarized in Table 3. Here we also compare the different variants of YOLOv5 and YOLOv8 for the n, s, m, and l models.

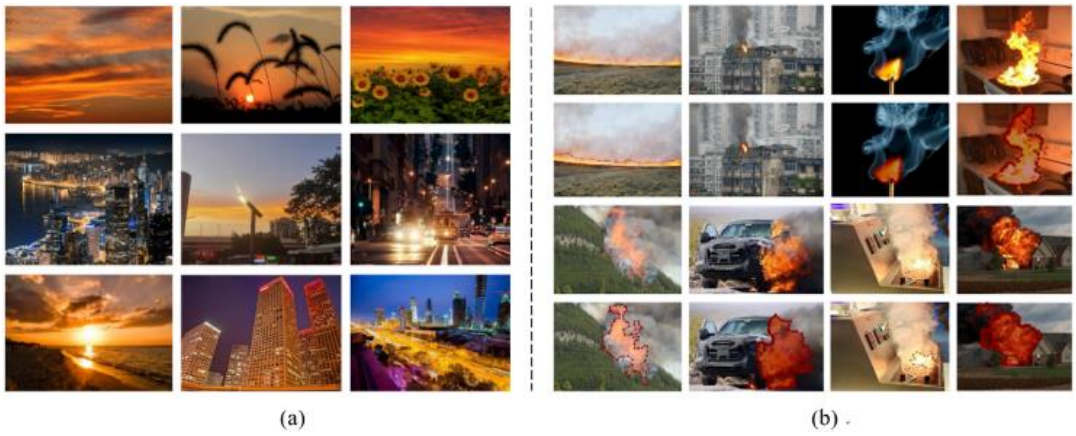


Fig. 2. Sample dataset. (a) Sample negative instance image. (b) Partial image of the dataset and annotation flow.

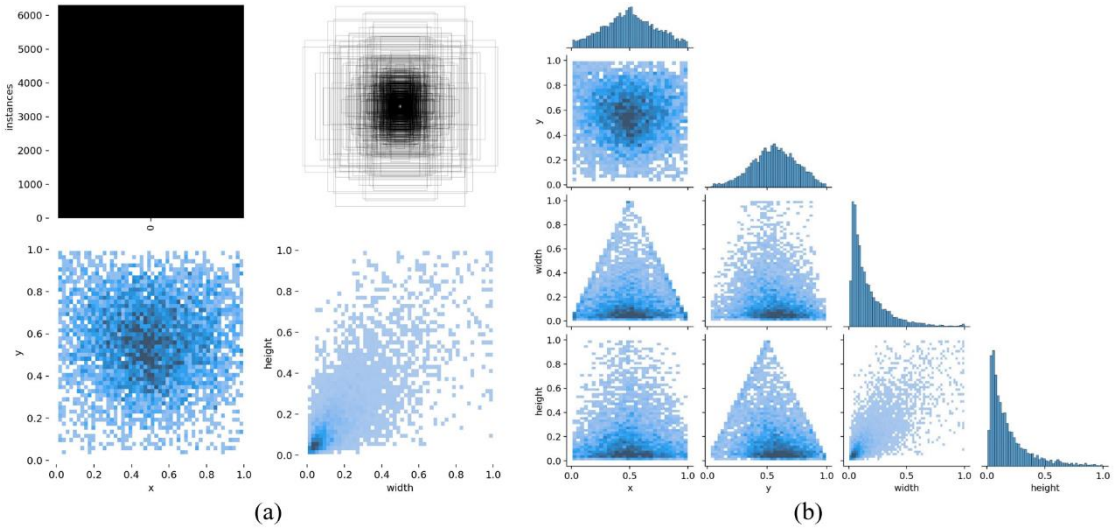


Fig. 3. Image dataset labels. (a) The visualization of size and spatial distribution. (b) Comprehensive visualization of label quantity, spatial distribution, and size in the dataset.

As shown in Table 3, the segmentation and detection precision of the proposed network reached 0.823 and 0.820, respectively, representing the best performance among the evaluated models. The precision for detection, segmentation, and the AP value of segmentation achieved by this network model are all optimal. Furthermore, compared to the original YOLOv7-tiny network model, the detection and segmentation precisions have been significantly improved by 3.8% and 3.6%, respectively. Although the F1(B), F1(M), and AP(B) metrics of this model are not the highest, they are only 0.2%, 0.4%, and 0.5% lower than those of the best-performing method, respectively. Additionally, compared to the original backbone network, the F1(M) and AP(M) values of the proposed network have been enhanced by 0.9% and 0.4%, respectively, demonstrating the best performance among the compared networks.

Fig. 4 presents the performance evaluation results of various attention mechanisms and feature extraction modules integrated into the YOLOv7-tiny network, compared to the original network structure model, with identical parameter settings. From left to right, the first eleven items represent different attention mechanisms and the original network model, while the last seven items represent various feature extraction modules. In Fig. 4, to facilitate a clearer comparison between F1(B) and F1(M) results, both primary and secondary axes are employed: the primary axis is on the left and the secondary axis is on the right, with F1(M) plotted against the secondary axis.

As illustrated in Fig. 4, the CBAM exhibits the most significant improvement in both box precision and mask precision, achieving values of 0.823 and 0.820, respectively, which demonstrates a clear advantage over other modules. Although CBAM does not show a marked advantage in the F1(B) and F1(M) metrics, the performance decline compared to other modules is minimal. Furthermore, the F1(M) score is still 0.9% higher than that of the YOLOv7-tiny network's baseline structure. Therefore, considering the comprehensive evaluation indices—P(B), P(M), F1(B), and F1(M)—the CBAM provides the most substantial overall improvement for the network structure.

Fig. 5 compares the computational complexity of the models incorporating different attention mechanisms using evaluation metrics such as Parameters, FPS, and FLOPS. As illustrated in the Fig. 5, while the CBAM module does not achieve the lowest parameter count among all models, it does not significantly increase the model's parameter count. Despite this, the CBAM module provides real-time segmentation with relatively low parameter occupancy and its FLOPS are comparable to those of the baseline model. Therefore, considering the evaluation metrics—P(B), P(M), F1(B), F1(M), Parameters, FPS, and FLOPS—the CBAM module is deemed optimal among all the evaluated modules.

4.1.4. Number of CBAM modules

This section presents the results of comparing network performance across different numbers of CBAM module replacements within the network structure. The aim is to determine the optimal number of CBAM module replacements by evaluating their impact on network performance under consistent hyperparameter settings. Specifically, the study involved substituting the simple convolutional layer with a 1×1 convolutional kernelfollowing the splicing layer in the MCB module of the backbone network. These configurations are referred to as 'M1', 'M2', and 'M3', corresponding to 1, 2, and 3 CBAM replacements, respectively.

Table 3

Comparison of optimal network(ours) and different network evaluation metrics.

Model	P(B)	P(M)	AP(B)	AP(M)	F1 (B)	F1 (M)
YOLOv5(n)	0.757	0.778	0.724	0.738	0.695	0.713
YOLOv5(s)	0.724	0.731	0.732	0.739	0.687	0.699
YOLOv5(m)	0.722	0.726	0.721	0.724	0.685	0.677
YOLOv5(l)	0.665	0.678	0.698	0.712	0.659	0.672
YOLOv7-seg	0.787	0.804	0.760	0.742	0.707	0.710
YOLOv7-tiny	0.785	0.784	0.773	0.769	0.748	0.739
YOLOv8(n)	0.793	0.802	0.723	0.723	0.692	0.702
YOLOv8(s)	0.742	0.799	0.724	0.737	0.703	0.719
YOLOv8(m)	0.732	0.738	0.735	0.738	0.707	0.709
YOLOv8(l)	0.806	0.816	0.743	0.756	0.717	0.727
YOLOv9(c)	0.790	0.804	0.758	0.767	0.748	0.752
Ours	0.823	0.820	0.768	0.773	0.746	0.748

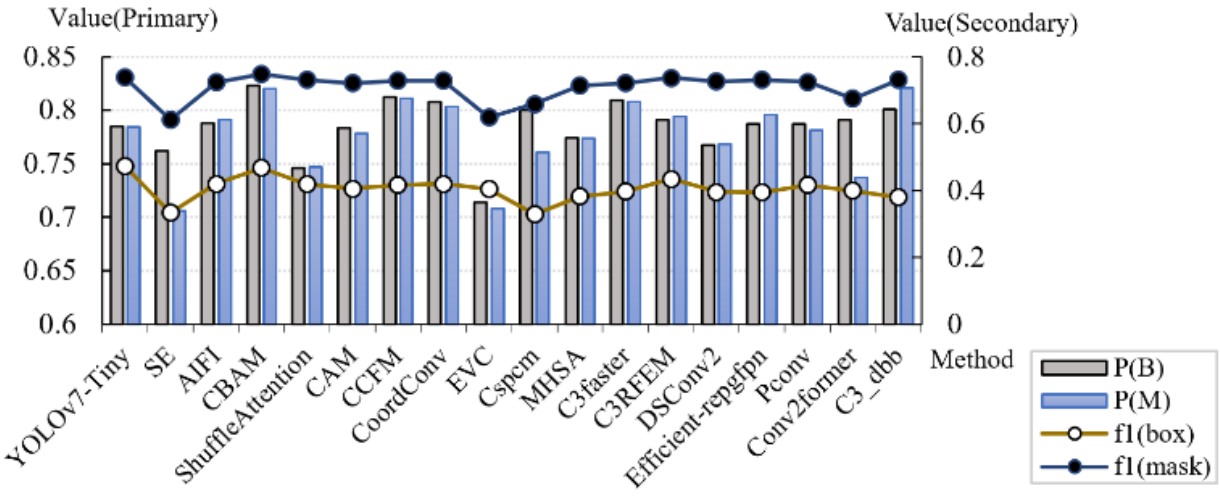


Fig. 4. Performance comparison of attention mechanisms and module placement in foundational network architecture.

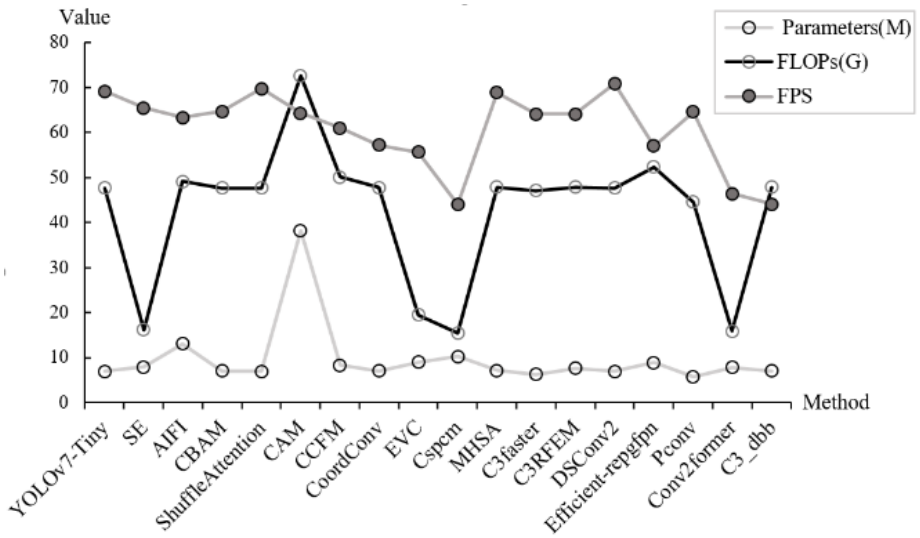


Fig. 5. Model size and inference time after placement of the attentional mechanism and feature extraction module into the model.

Table 4 presents the performance comparison results between the three CBAM module replacement methods and the original baseline model. The results indicate that the 'M2' method exhibits the best overall performance. Specifically, the 'M2' method demonstrates a 3.8% improvement in box precision and a 3.6% improvement in mask precision compared to the baseline model, making it the optimal replacement method for both metrics. Additionally, 'M2' shows a competitive edge in F1-scores, with a 3.2% increase over the baseline model in box precision and a 2.4% increase over the second-best method. For mask precision, 'M2' achieves a 5.0% improvement over the baseline model and a 2.3% increase over the second-best method, further underscoring its effectiveness. In terms of AP values, 'M2' is also superior, with a 0.4% increase in the AP(M) value compared to the benchmark model. Although the AP(B) value is slightly lower, it is only 0.5% below the highest benchmark. The 'M2' method yields an FPS value of 64.724, meeting the real-time segmentation requirements. Moreover, 'M2' results in only 10,820 more parameters than the original model while maintaining superiority in precision and F1-score. Considering these factors, the 'M2' method is deemed optimal and has been adopted in the network model for this research.

Table 4

Comparative network performance results for different numbers of CBAM attention mechanisms.

Method	P(B) (↑)	P(M) (↑)	AP(B) (↑)	AP(M) (↑)	F1 (B) (↑)	F1 (M) (↑)	FPS(↓)	Params(↓)
original	0.785	0.784	0.773	0.769	0.714	0.698	69.140	6987372
M1	0.779	0.782	0.747	0.739	0.722	0.725	69.416	6989646
M2(ours)	0.823	0.820	0.768	0.773	0.746	0.748	64.724	6998192
M3	0.783	0.794	0.734	0.732	0.701	0.708	60.04	7031570

4.1.5 Inference time and model size

In order to better examine the computational complexity and spatial complexity of the models, we set the batch size to 1 and compare the inference speeds of all the network models in this research, as shown in Table 5. Here, FPS is used to measure the processing speed of the models for real-time images and videos. This represents the number of frames processed per second. When the FPS value is greater than 30, the requirement of real-time segmentation is satisfied. The evaluation results of FLOPs, Params, and FPS show that our model achieves real-time segmentation with smaller parameter occupancy and realizes high-precision detection and segmentation of wildfires compared to these existing methods, and the model in this research stands out as a lightweight but effective solution.

3.2. Wildfire front extraction based on region-locking

4.2.1. Description of the outdoor prairie fire fusion ignition experiment

An ignition experiment was conducted on March 29, 2023, at Cao Tang Gully in the west mountain of the Laoshan Experimental Station of the Maor Mountain Experimental Forestry Farm of the Northeast Forestry University in Shangzhi, Harbin, China. The ignition time was 10:54:55 a.m., with a burning duration of 2 minutes and 10 seconds, covering a region of 0.01 hectares. Low shrubs (≤1 m in height) on the horizontal ground were selected for the experiment. The ignition region of the experimental sample plot was 375 m². Its aerial view and conceptual shape diagram are shown in Fig. 6 (a) and (b), respectively. The location circled by the red box in Fig. 6 (a) is the general location of the experimental sample plot, and the yellow triangle is the location of the specific shape of the experimental sample plot. Fig. 6 (b) demonstrates the specific shape of the experimental sample plot, which is an isosceles triangle △ABC with a base side of 20 cm, a height of 37.5 cm, and an apex angle of 30 °. Before the experimental burning, the experimenters pruned the vegetation within the experimental plots to achieve the desired isosceles triangular shape, and a 15-meter-wide barrier strip was set up around the experimental region of the ignition experiment.

Table 5

FPS, FLOPS and Params's comparison results between the optimal network (ours) and different networks.

Model	FLOPS/G(↓)	Params/M(↓)	FPS(↑)
YOLOv5(n)	6.70	1.88	79.10
YOLOv5(s)	25.70	7.40	73.43
YOLOv5(m)	69.8	21.65	63.14
YOLOv5(l)	147.00	47.50	20.00
YOLOv7-seg	141.90	37.84	53.21
YOLOv7-tiny	47.70	6.99	69.14
YOLOv8(n)	12.00	3.26	50.01
YOLOv8(s)	42.40	11.78	45.98
YOLOv8(m)	110.00	27.22	42.00
YOLOv8(l)	220.10	45.91	25.32
YOLOv9(c)	144.20	27.36	32.54
Ours	47.70	7.00	64.72

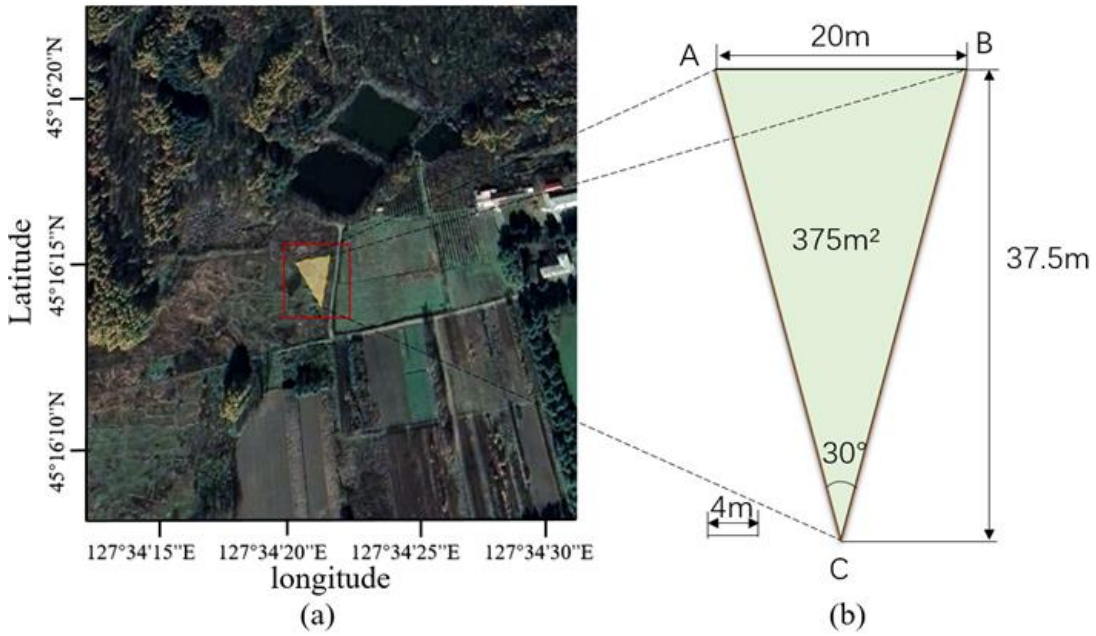


Fig. 6. Aerial view of the experimental sample plot with schematic shape structure. (a) Aerial view of the experimental sample plot ("Google Earth," 2021). (b) Schematic diagram of the shape structure of the experimental sample plot.

Before the ignition experiment began, we took 5 grams of surface fine dead combustibles from 3~5 random sample sites for moisture content determination, and the specific values of the moisture content of the surface fine dead combustibles from the random sample sites are shown in Table 6. We took 3~5 random sample sites from the sample sites and used portable meteorological meters to determine the ambient temperature, relative humidity, wind speed, etc. We recorded the changes in environmental factors before and after the ignition and used the Portable meteorological instrument Kestrel 4200 Pocket Air Flow Tracker to determine the ambient air temperature, relative humidity, wind speed, etc. at a distance of 8m from the apex of the isosceles triangle fire field and a height of 1.6m from the ground. The specific values of the ambient factors are shown in Table 7.

Table 6

The water content of surface fine dead combustibles was randomly sampled.

Fine surface fuels	Moisture content /5g
Sample 1	13.33
Sample2	6.44
Sample3	6.28
Sample4	7.71
Sample5	7.46
Mean	8.24

Table 7

Changes in ambient factors before and after ignition.

time	Temperature (℃)	humidity (%)	wind speed (m·s-1)	wind direction
Before ignition	14	29.4	2.5	Southwest
1 minutes after ignition	14.8	30.3	1.3	Southwest
2 minutes after ignition	15.8	30.9	0.8	Southwest
3 minutes after ignition	13.9	30.3	2.0	Southwest
4 minutes after ignition	14.6	31.0	2.0	Southwest

After the preparations were completed the ignition was carried out. The experimental ignition utilized a cotton string soaked with gasoline. Specifically, the experimenter ignited the fire simultaneously on both sides of the isosceles triangle experimental sample plot (positions AC and BC of the orange-red line in Fig. 6(b)), and to ensure that the two fires were completed at the same time, multi-point simultaneous ignition was utilized. Before the experimental combustion, the UAV, modeled as DJI M600, was directed by ground command and control to fly to the airspace above the experimental fire field. High-speed photography was used to record the entire fire spreading process of the experimental sample fire field and to obtain the RGB images of the fire front spreading in the experimental sample field.

4.2.2. Implementation details of fire field front extraction

The RGB images capturing the ignition process of the experimental fire field, acquired by the DJI M600 UAV during the outdoor prairie fire fusion ignition experiment described in the previous subsection, serve as the data source for wildfire front extraction in this subsection. Due to the fire terrain, wind speed, wind direction, temperature and humidity, and other ambient factors, the fire spread process may manifest in different regions: the fire region, burned region, and unburned region. The fire spread may also occur intermittently. The temperature at the fire front is extremely high, and close contact can be life-threatening. Therefore, we define the fire front as the leading edge of the entire fire spread, meaning the foremost part of the fire during its ignition and spread., details are provided in Fig. 7.

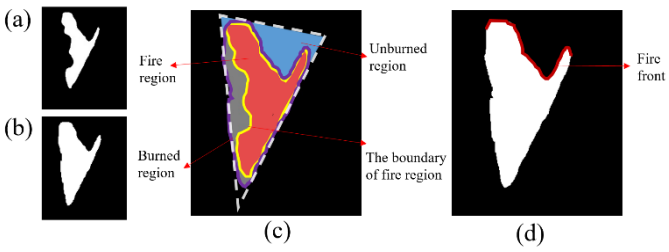


Fig. 7. Schematic diagram of fire region boundary and fire field division. (a) The fire region map at the current moment. (b) The fire region time accumulation map at the current moment. (c) Schematic diagram of fire field delineation and fire region boundary. (d) Schematic diagram of the fire front.

Fig. 7 presents various aspects of the fire field: Fig. 7(a) shows the fire region map at the current moment; Fig. 7(b) depicts the time-accumulated fire region map at the current moment; Fig. 7(c) provides a

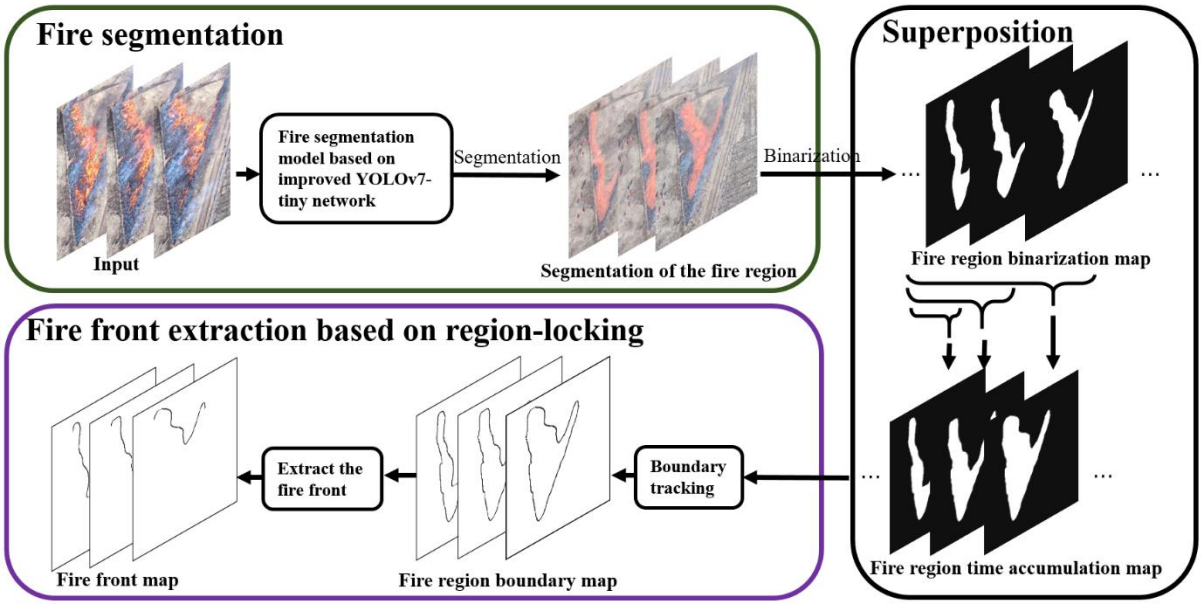


Fig. 8. Architecture diagram of region-locked wildfire front extraction algorithm.

schematic of fire region delineation and boundary; and Fig. 7(d) illustrates the fire front. In Fig. 7(c), the red region represents the fire region, the blue region denotes the unburned region, and the region enclosed by the purple line indicates the burned region (comprising both the red and gray regions). The red line in Fig. 7(c) marks the fire front, defined as the boundary between the fire region and the unburned region.

We first performed preprocessing operations on those RGB image sequences, including image registration and image enhancement, to obtain high-quality preprocessed images of the experimental plots. Using these preprocessed images, we then employed a trained and optimized fire segmentation model to accurately segment the fire regions. Next, the image is binarized to obtain a binarized map of the fire region. Subsequently, for each frame, we accumulated and superimposed the binary images of the current and previous frames in the time sequence to obtain the fire region time accumulation map. Then the time accumulation map is used to interpret the fire front through the region-locked fire interpretation algorithm to obtain the fire front of the fire field in the experimental sample site. Specific realization steps are shown in Fig. 8.

Specifically, due to the change of flight attitude during UAV shooting and the spatial offset and rotation of the image brought about by high-speed photography, which will lead to certain geometric distortion among the frames in the image sequence, we need to preprocess the RGB image sequence of fire spreading in the experimental sample site acquired by the UAV DJI M600, which includes the steps such as image registration and image enhancement. Using the optimized fire segmentation model obtained through multiple generations of iterative training, we segmented the preprocessed image data to extract the fire regions in the images. Then, the obtained segmented image is binarized to obtain the fire region mask map. Since the fire front is defined in this research as the forefront of the entire spreading process of fire, we superimpose each frame of the fire region binarization map by time series to obtain the fire region time accumulation map at each moment. Here, the time unit of each frame is second, and the time accumulation image of each moment is the superposition of the image of the current moment as well as the previous moment. According to the fire region time accumulation map of each moment, boundary tracking is performed in accordance with boundary tracking rules to obtain the spreading boundary of the fire field, and then the boundary map of the fire field of each moment is obtained. Based on the fire boundary map, the fire front is extracted from the region of interest based on region locking, and the fire front of the entire spreading process of the experimental sample site is finally obtained.

4.2.3. Experimental results and analysis

In order to demonstrate the effectiveness of the region-locking-based wildfire front extraction algorithm proposed in this research, we conducted a series of experiments based on the image data from the outdoor prairie fire fusion ignition experiment. We first carried out preprocessing work on the extracted image data of the outdoor prairie fire fusion ignition experiment, including operations such as image registration and image enhancement. The partial sample of the pre-processed experimental image data is shown in Fig. 9.

Fig. 10 shows the pre-processed fire images and segmentation effect of Cao Tang Gully in the west mountain of the Laoshan Experimental Station of the Maor Mountain Experimental Forestry Farm of the Northeast Forestry University in Shangzhi, Harbin, China, on March 29, 2023, from 10:56:00 to 10:56:32 BST. Here, due to the space problem, some images are taken to show the segmentation effect of the fire segmentation model proposed in this research. The first row of Fig. 9 is the preprocessed image; the second row is the fire region segmentation map. As can be seen from the Fig. 10, the fire segmentation model in this research accurately segments the fire in the image and proves its reliability.

Fig. 11 shows the fire region time accumulation and boundary map, the first row is the fire region binarization map, the second row is the fire region time accumulation map, and the third row is the fire region time accumulation boundary map. From the Fig. 11, we can see that this paper uses the fire region binarization map to accumulate according to time and obtains the fire region time accumulation map at each moment. This research adopts boundary tracking to obtain the outline of the fire field and traces the complete boundary of the fire field according to the topological structure relationship between the boundary pixels, and the obtained boundary of the fire field is accurate and continuous, which provides the premise for the subsequent deciphering of the fire front of the fire field, and proves the boundary tracking method chosen in this paper is effective.

Fig. 12 is a fire field interpretation map, the first row is a fire field reconstruction map, the second row is a region of interest map of the fire region time accumulation boundary, and the third row is an interpreted fire front map, where the red triangular box in the second row is the region of interest. The gray plus white color in the fire field reconstruction map indicates the burned region, the white color indicates the fire region at the current moment, and the red closed line segment indicates the fire region boundary at the current moment. It can be seen from the fire reconstruction map of the fire field that the boundary of the fire region at the current moment does not represent the forefront of the whole fire field, so the fire region time accumulation map is needed to obtain the fire front. The fire front we extracted from the region of interest based on the fire region time accumulation boundary is accurate and continuous, which paves the way for further work in the future to decipher detailed information such as fire

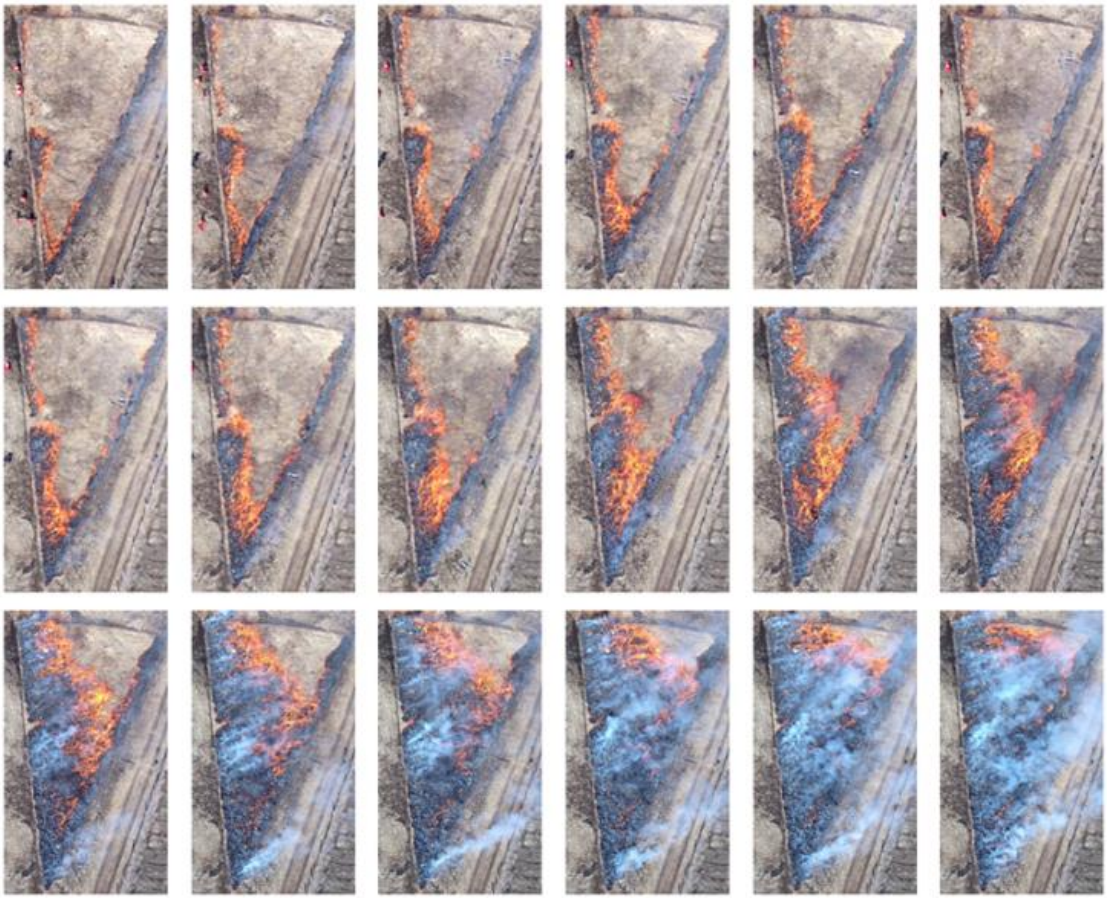


Fig. 9. Partial sample of preprocessed image data from outdoor prairie fire fusion experiment.

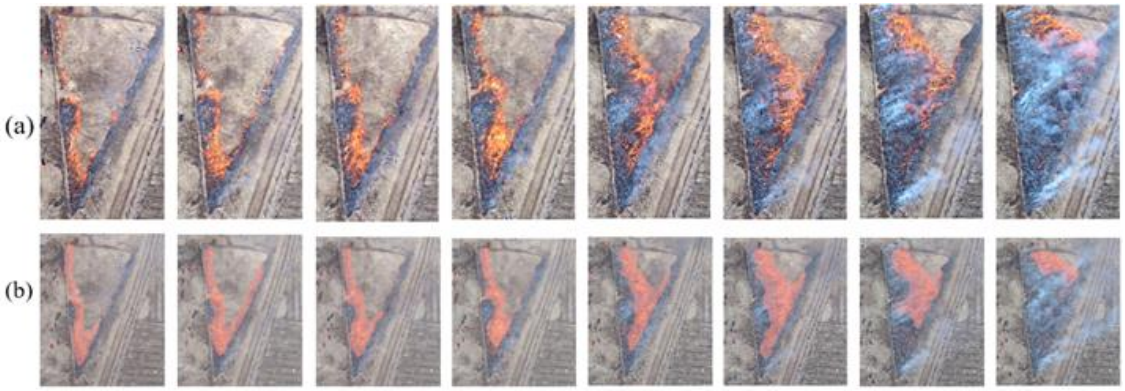


Fig. 10. Segmentation result of the fire region. (a) Preprocessed image. (b) Segmentation of the fire region.

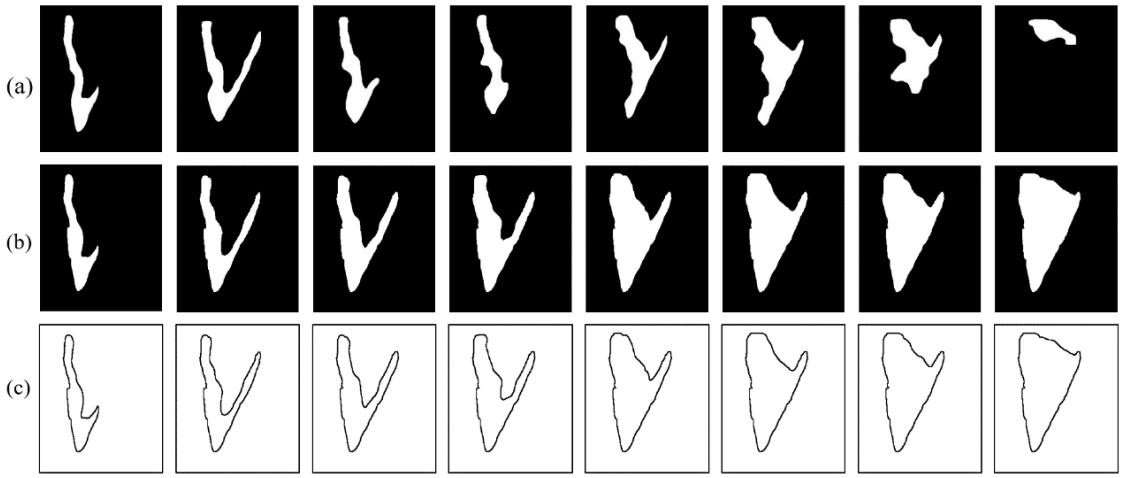


Fig. 11. Fire region time accumulation and boundary map. (a) Fire region binarization map. (b) Fire region time accumulation map. (c) Fire region time accumulation boundary map.

behavior (fire spreading speed, fire spreading direction and fire intensity, etc.) of the fire field.

Fig. 13 shows the general view of the fire spread at the fire field, Fig. 13(a) is the fire spread map, and Fig. 13(b) is the overall map of the fire

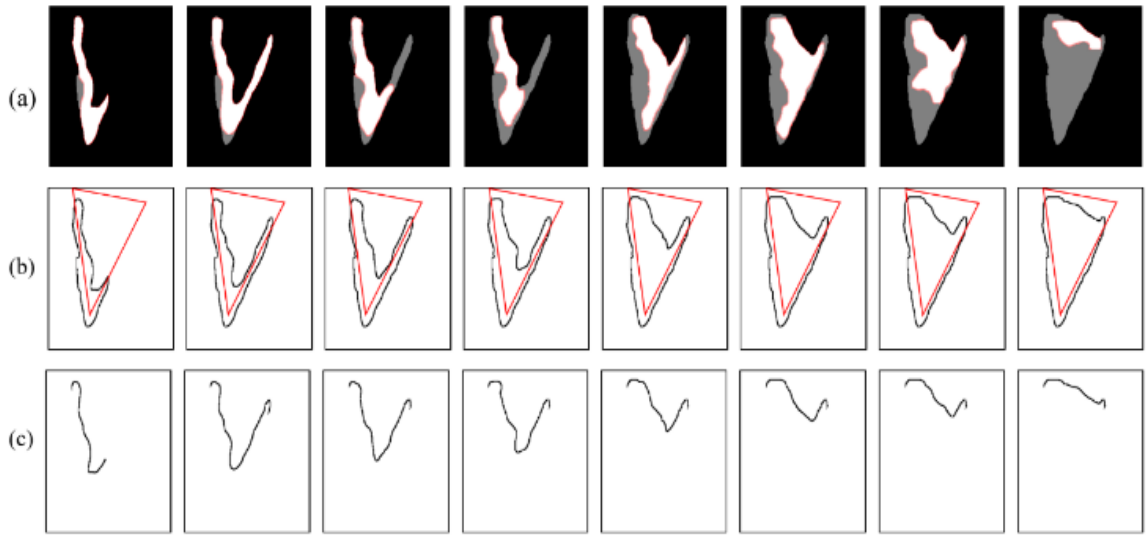


Fig. 12. Fire field front interpretation map. (a) Fire field reconstruction map. (b) Region of interest map of the fire region time accumulation boundary. (c) Interpreted fire front map.

field front. Here we randomly selected some moments of the fire field front to show the whole fire field spreading process and used different colors to indicate the fire spreading growth region and fire field front in the corresponding moments. From Fig. 13(a), we can clearly see the fire spread growth region of each moment compared to its previous moment, and here the fire spread growth region is represented by different colors. Fig. 13(b) uses different colors to represent the fire front of the corresponding moment. From Fig. 13, it can be seen that this outdoor prairie fire experiment was conducted under southwest wind conditions, the fire spread from the southwest of the triangular experimental region to the northeast. In addition, the FPS of this experimental image segmentation reaches 43.29 HZ, and the Precision of segmentation reaches 82.4%, which meets the real-time segmentation requirements and realizes high-precision segmentation.

Fifty-eight field ignition experiments were conducted in Victoria and Queensland, Australia, by Miguel G. Cruz and colleagues from the Commonwealth Scientific and Industrial Research Organisation (CSIRO), in collaboration with the Australian National Fire Authority (CFA) (Cruz et al., 2018). To demonstrate the robustness of the fire interpretation algorithm proposed in this study, we also performed fire interpreting for Experiment E2 and Experiment E21 conducted by Miguel G. Cruz et al (2018) in the town of Braidwood, southern New South Wales, Australia. Both experiments were ignited on December 18, 2015, at 10:38:00AM, with a burn duration of 36 minutes and a rectangular test site shape. The relative humidity, temperature (Degrees Celsius), and wind speed of the experimental environment were 35%, 27°C, and 25km/h, respectively, in addition to the fuel load (fuel load) of 2.6T/HA for the Experiment E21 and 2.1T/HA for Experiment E2. Specific details of the experiments are described in detail in the literature (Cruz et al., 2018).

Fig. 14 shows the overall fire spread map of the experiments on the test site in the town of Braidwood, southern New South Wales, Australia, which were obtained through fire interpretation, Fig. 14(a) and (b) are the fire spread and fire front maps of Experiment E21, and Fig. 14(c) and (d) are the experimental fire spread and fire front maps of Experiment E2, respectively, where we have selected some moments to show the spreading process of the two experiments. It can be seen that the model of this study successfully interprets its fire field front, which proves the robustness of the model. In addition, the FPS of image segmentation in Experiment E2 and Experiment E21 reached 45.66 HZ and 42.19 HZ respectively, and the segmentation precision reached 86.4% and 85.2% respectively, which both

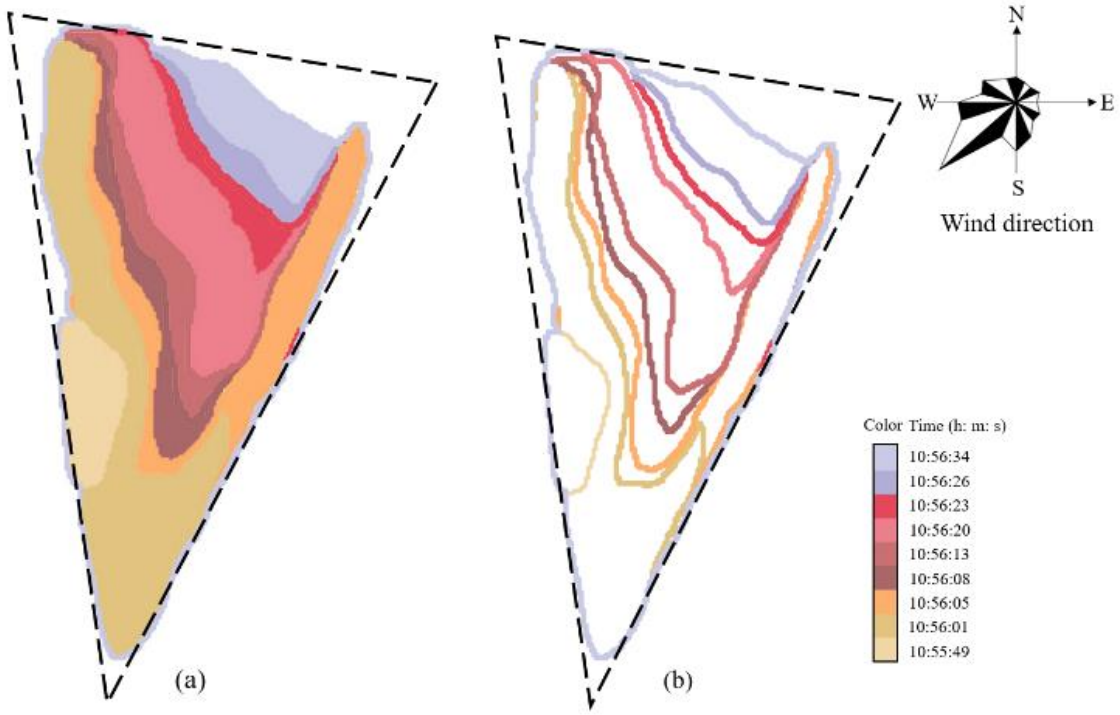


Fig. 13. General view of the fire spread at the fire field. (a) Fire spread map. (b) Fire front at the fire field.

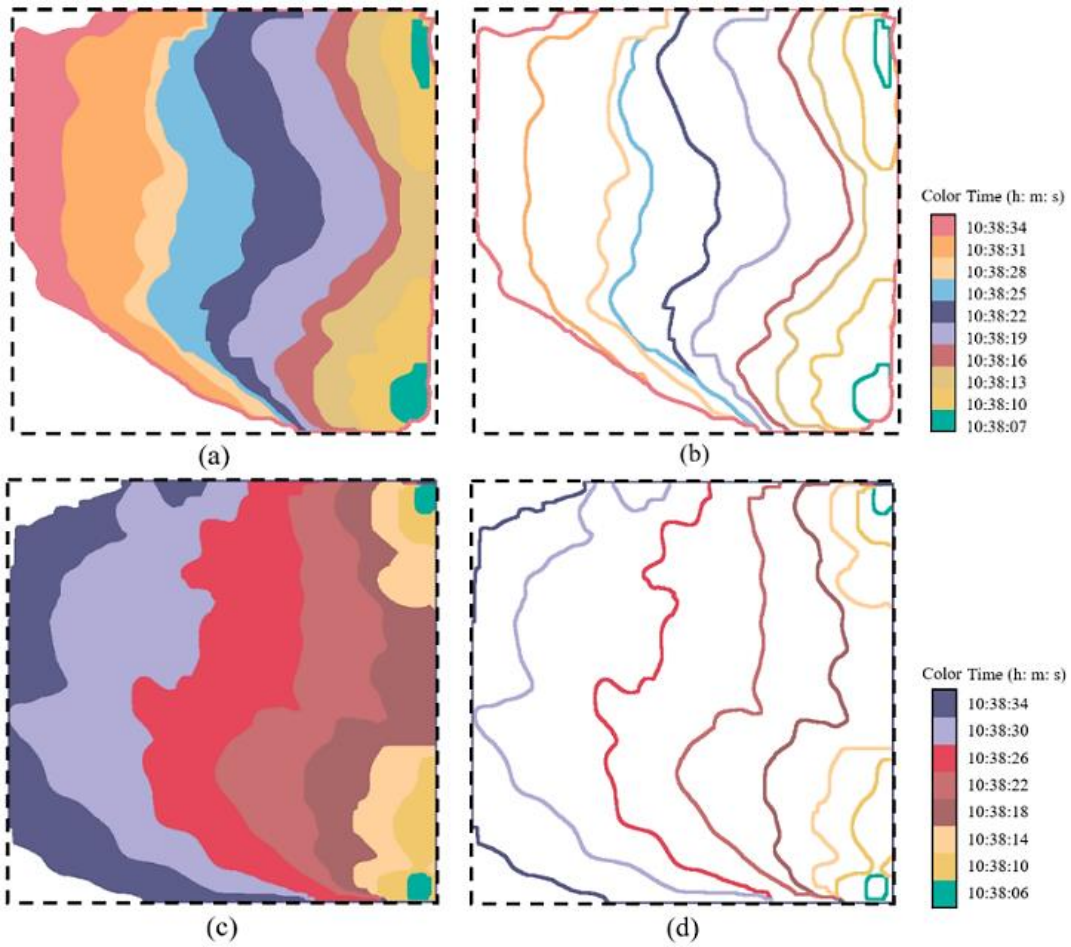


Fig. 14. General view of the fire spread at the fire field for Experiment E2 and Experiment E21. (a) Fire spread map for Experiment E21. (b) Fire front at the fire field for Experiment E21. (c) Fire spread map for Experiment E2. (d) Fire front at the fire field for Experiment E2.

achieved real-time segmentation effect and realized high-precision segmentation.

5. Conclusion and discussion

The focus of this research is on the intelligent interpretation of fire front in a fire field to provide information on details such as fire front, and to do preliminary work for further extraction of fire behavior in the future. There are challenges in wildfire front extraction based on UAV visible light images, and we learned that the current fire front extraction methods mainly extract fire front by inverting fire front through thermal sensors, predicting fire front based on wildfire spread models, and searching for fire front based on thermal image thresholds, and there is little research work on interpreting fire front based on UAV visible light images combined with traditional mathematical algorithms and deep learning algorithms.

To solve these fundamental problems, our research makes the following contributions:

- We proposed an intelligent UAV low altitude prairie fire front interpretation method based on fire segmentation region-locking for the first time, which accurately extracts fire front from fire field images by combining the fire segmentation model and region-locking based wildfire front extraction algorithms, and realizes the intelligent interpretation of wildfire front based on UAV visible light images.
- We proposed a lightweight fire segmentation model that improves the existing YOLOv7-tiny network, focuses the model's attention on the fire features by adding the CBAM to make the model better segment the fire, and learns the main feature information of the fire through iterative training in multiple generations to realize the effect of fire region segmentation in the fire field.
- We proposed a wildfire front extraction algorithm based on region-locking, which obtains the fire region boundary information through boundary tracking, performs region-locking based on the fire region boundary information, and intelligently interprets the fire front for the region of interest.

Based on the above insights, although the methodology of this research has better performance, certain limitations must be recognized, which can be focused on and improved in future research work:

- The dataset can be further improved towards labeling quality and scene diversity. The labeling quality affects the learning capacity of the fire segmentation model, so the labeling quality helps to improve the model segmentation precision. In addition, the dataset scene diversity can be increased, such as by appropriately adding images of foggy days, light smoke, and blurred fires, which can enhance the model's generalization ability.
- The network structure of the wildfire segmentation model can be further optimized. The improved network model derived from this research is the optimal solution obtained from the experiments in this research, and in the future, the network structure of the model can be further optimized for the attention module, convolutional layer, and loss function to enhance segmentation performance.
- The extraction scheme of the fire front of the fire field can be further optimized. In this research, fire front extraction is realized by a wildfire front extraction algorithm based on region-locking, which needs to determine the region of interest, and future related research can be directed towards how to formulate a rule to intelligently determine the region of interest of a fire, making the algorithm more intelligent.
- Since this research model is based on UAV images for detection and segmentation, UAV images with obstacles that obscure the fires will

affect the detection and segmentation effect of the model, so this research model is more suitable to be applied to fire scenarios (prairie fires) where the terrain is more gentle and the field of view is more open. In more complex real-world conditions, such as dense vegetation or uneven terrain, multiple UAVs may be deployed to capture images. By adjusting their altitude and orientation, the UAVs can obtain comprehensive visual data of the entire fire field, thereby improving the model's detection and segmentation performance. Future research can consider the terrain and vegetation issues so that the model can be applied to a wider range of scenarios.

The proposed method has been tested based on image data from actual wildfire experimental sample plots, which confirms the effectiveness of the method. The next step of the work proposes to optimize the fire segmentation model in future research to further improve the precision of fire segmentation. In addition, the wildfire front extraction scheme will be further optimized to make wildfire front extraction more intelligent and easy, and the results will be applied to the development of dynamic data-driven forest fire spread models and digital twinning of fire fields.

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